

VSC Simulations: What We Did

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September 29, 2026

Overview

We ran cavity Born–Oppenheimer MD simulations of single molecules inside optical microcavities using `polaritonic_deep_md` (Flick group). When the cavity frequency matches a molecular vibration, the IR peak splits into a lower polariton (LP) and upper polariton (UP). The splitting $\Omega_R = \text{UP} - \text{LP}$ is the Rabi splitting.

CO₂

We used CO₂ as a benchmark because a trained neural network potential (NEP) exists, making runs fast. We tested three force backends (FF, PySCF, NEP) and confirmed NEP reproduces DFT-quality dipoles.

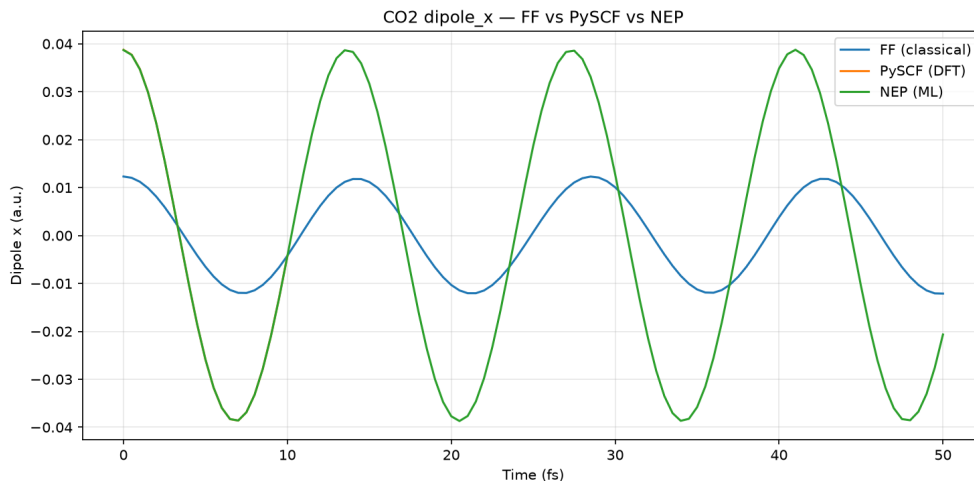


Figure 1: CO₂ dipole over time. FF gives smaller amplitude; PySCF and NEP are nearly identical.

We then ran at coupling strengths $\lambda = 0, 0.1, 0.2, 0.3$ reproducing Figure 1 of Bonini et al. 2024. At $\lambda = 0.1$ the asymmetric stretch ($\sim 2349 \text{ cm}^{-1}$) splits with $\Omega_R \approx 420 \text{ cm}^{-1}$.

H₂O

We ran water using PySCF only (no NEP model exists for H₂O). The cavity was tuned to the bend mode at 1584 cm⁻¹, polarisation along $\hat{z}$. 2000 steps at 0.5 fs timestep, Hanning-windowed FFT.

Lower polariton (LP)	1517 cm ⁻¹
Upper polariton (UP)	1642 cm ⁻¹
Rabi splitting	125 cm⁻¹

H₂O splits less than CO₂ (125 vs 420 cm⁻¹) because the water bend has a smaller dipole derivative than the CO₂ asymmetric stretch.

Next: liquid water with LAMMPS — 216 molecules, collective splitting scales as $\sqrt{N}$, so $\sim 15\times$ the single-molecule result.

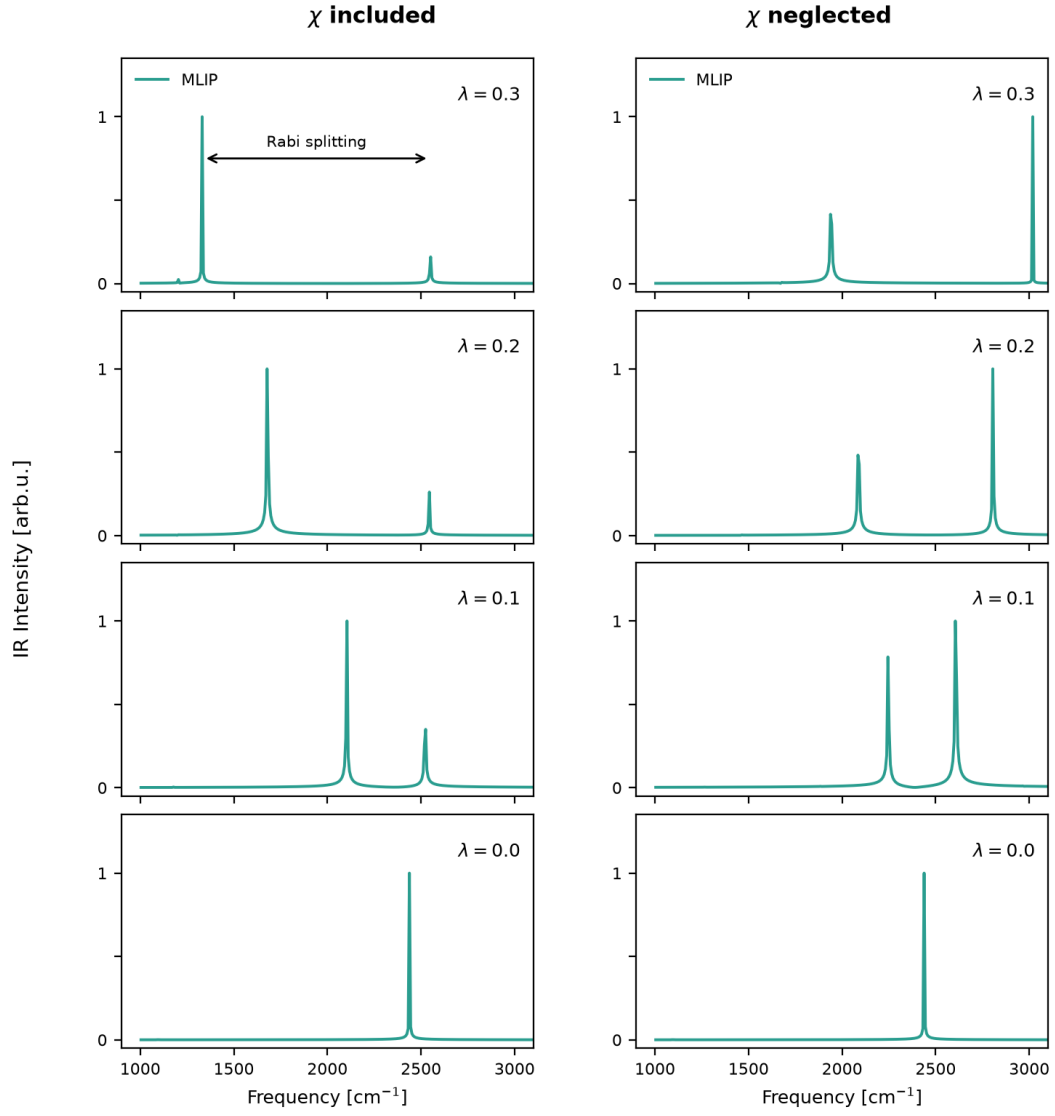


Figure 2: Stacked IR spectra of CO₂ at four coupling strengths. The peak splits into LP and UP as λ increases. Matches Bonini et al. 2024.

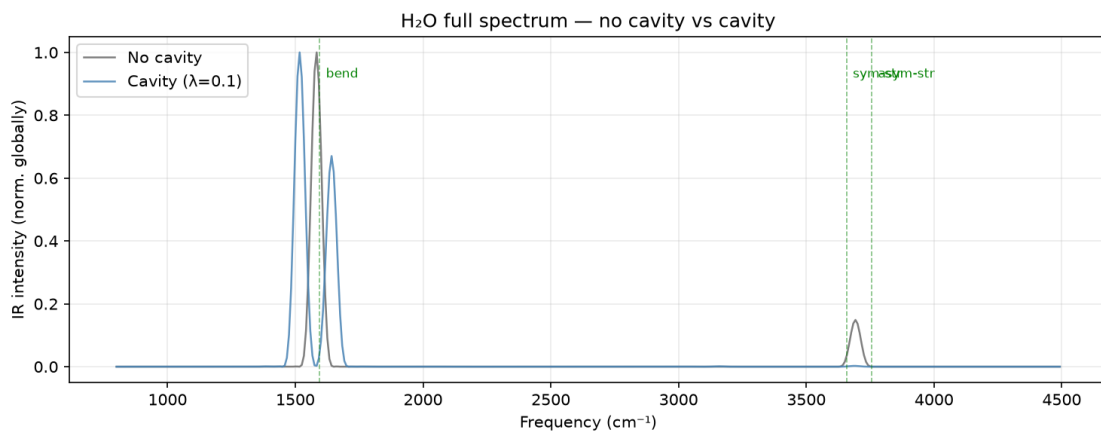


Figure 3: Full H₂O spectrum: no cavity (grey) vs cavity (blue). The bend splits in the cavity run. The O–H stretch at 3657 cm⁻¹ is unaffected — cavity coupling is frequency-selective.

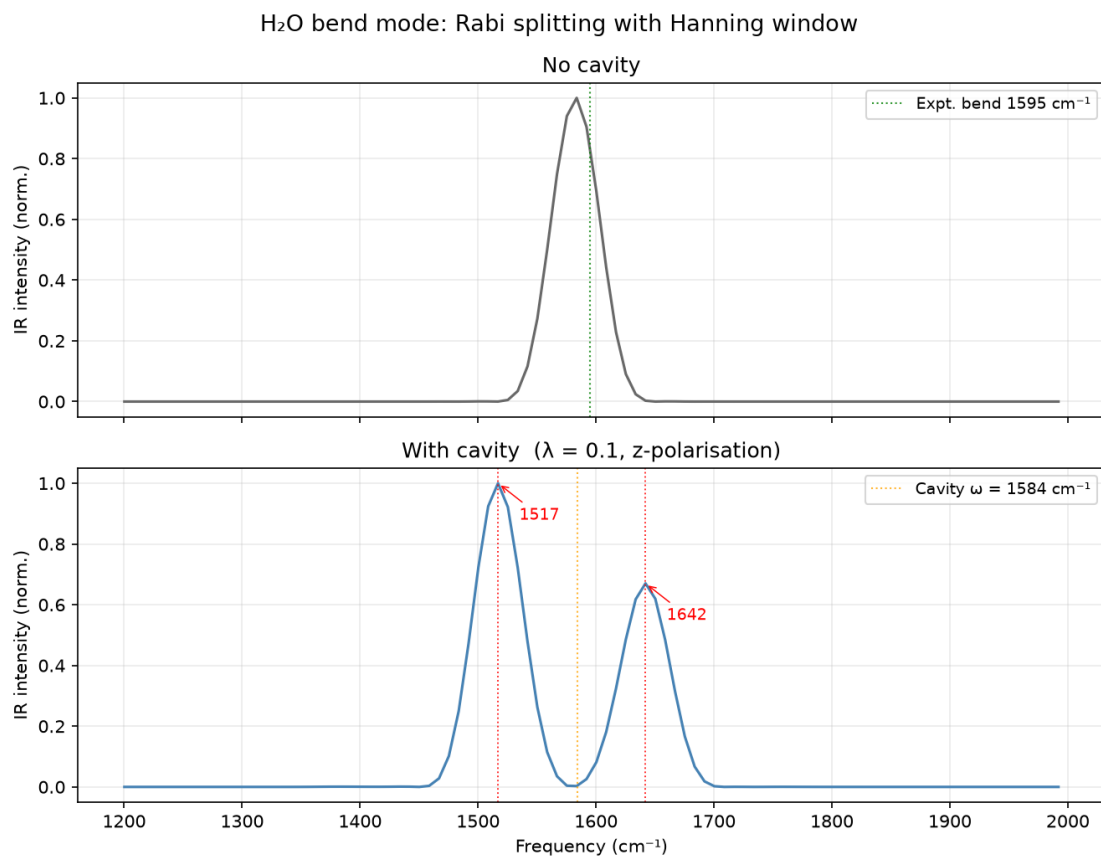


Figure 4: Bend region in detail. Top: bare molecule, single peak at ~ 1580 cm⁻¹. Bottom: cavity run showing clear LP/UP splitting.